The Row Sums Were the Method, Not the Theorem: a Machine-Checked Dobrushin Resolvent Chain, and a Misattributed Uniformity Wall

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Abstract

A formal development of a two-dimensional spatial transfer kernel proves its uniform spectral statements by Schur's test on *constant* row sums, and carries a theorem, labelled THE OBSTRUCTION in its own source, stating that those row sums stop being constant the moment the spatial coupling is switched on. That theorem obstructs the *method*. It was read as obstructing the *conclusion*, and the reading outlived its own retraction.

We prove, in Lean 4 with no `sorry` and no project axiom, a chain of three theorems. **First**, the one-bond influence envelope: flipping one neighbour across a bond of strength $J \geq 0$ moves a two-state site's conditional by at most $\tanh J$, and by exactly $\tanh J$ at one field, so $\tanh J$ is the *least field-uniform* bound. It is an interdependence majorant, not the minimal coefficient of any particular finite model, whose reachable fields form a discrete set — a distinction Section 2 makes precisely. **Second**, those envelopes assembled into a matrix discharge every *matrix-side* hypothesis of the third (the two conditions on the ambient distance remain hypotheses), and turn a purely *local* covering condition into the remaining row-sum bound holding at every site of every volume — which is the coupling window, and which we compute: at a site carrying two bonds of strength β and two of strength γ it is exactly $2\tanh|\beta| + 2\tanh|\gamma|$, so the window is the hypothesis of a theorem rather than a sentence. **Third**, the resolvent bound: for a nonnegative matrix C over a finite index type, supported on pairs at distance at most one and with row sums *bounded* by $\alpha < 1$ and never constant,

$$\sum_{n<N} (C^n)_{ij} \;\leq\; \frac{\alpha^{\mathrm{dist}(i,j)}}{1-\alpha} \qquad \text{for every } N,$$

with no reference of any kind to the cardinality of the index type. Each rung carries a non-vacuity witness and every declaration is oracled.

Separately, and by measurement rather than proof, we remove the wall from where it was being placed. For the coupled kernel the second-eigenvalue ratio specRatio(L) saturates strictly below one in each of six cells tested inside Onsager's disordered region — with the spatial weight switched on — and reaches one to six decimal places in the two ordered cells tested. The control at zero spatial coupling reproduces the *proved* identity specRatio $=$ $\tanh\beta$ to six decimals, so the harness is calibrated against a theorem rather than against itself. These data are eight cells at $L \leq 12$ with an extrapolation. On the tested grid they *contradict* the blanket attribution of the degeneracy to the mere presence of the spatial weight; they are not, and are not offered as, a refutation of an asymptotic claim, since a sequence can appear to saturate at $L \leq 12$ and turn afterwards. They do *not* locate the transition, which would need a dense sweep and finite-size scaling that we do not attempt.

What is not proved. The volume-uniform bound on specRatio for the coupled kernel is *not* established here. Dobrushin's uniqueness theorem is classical [1, 3]; nothing in this paper is new mathematics. What is offered is a mechanised, non-vacuous chain, measured

counterevidence against an attribution, and a single remaining obligation named exactly: Dobrushin's comparison estimate. No consequence for Yang–Mills theory is stated, suggested, or implied.

1 What this paper claims, and what it does not

Claimed, machine-checked. Theorems 3.1, 3.2 and 3.3, their witnesses and the corollary that composes them, in `DobrushinCoefficient.lean`, `DobrushinRowSum.lean` and `DobrushinMatrix.lean`. Every declaration is oracled and depends on exactly Lean's three standard axioms.

Claimed, measured. The saturation table of Section 4. These are floating-point computations at finite extent with a documented extrapolation; they are evidence, not proof, and they are labelled as such wherever they appear.

Claimed, about the record. Section 2 states precisely what the source theorem `coupled_rowSums_not_constant` does and does not license. This is a statement about a formal development, and it is verifiable by reading that development.

Not claimed. That specRatio of the coupled kernel is bounded away from one uniformly in the extent. That the window derived from Dobrushin's condition is optimal — it is not, and Section 6 says by how much. That any of this bears on the Yang–Mills mass gap.

2 The setting, and the theorem that was read too widely

Fix an extent L and consider configurations $\sigma \in \{0,1\}^L$ of one slice. The development under discussion carries a time-bond kernel that factorises over sites,

$$K_\beta(\sigma,\tau) \;=\; \prod_{j<L} b_\beta(\sigma_j,\tau_j), \qquad b_\beta(s,s') = e^{\pm\beta},$$

and an arbitrary strictly positive slice weight w, the Gibbs weight of a space-time configuration being the product of the slice weights and the time bonds. When w is itself a product over spatial bonds with coupling γ, the system is the anisotropic two-dimensional Ising model.

Two facts from that development matter here.

Proposition 2.1 (proved elsewhere in the development)**.** *At $\gamma = 0$ and for every $L \geq 1$,* $\mathrm{specRatio} = \tanh\beta$ *exactly.*

The proof is Schur's test [6]: the decoupled kernel has *constant* row sums, and the loss on the odd part of an observable is exactly compensated by the odd bond eigenvalue at every step.

Proposition 2.2 (`coupled_rowSums_not_constant`)**.** *At $\gamma > 0$ the row sums of the coupled kernel are not constant: two explicit configurations of a two-site slice have different row sums.*

Proposition 2.2 is labelled THE OBSTRUCTION in the source. *It obstructs Schur's test.* It says nothing whatever about specRatio, it is proved on a two-site witness, and no statement quantified over L appears in it.

A stronger reading — that the spatial weight destroys uniformity — was entered into the development and then *retracted of record* as unproved, leaving behind only a measurement of degeneracy.

What is documented, and can be checked. The retraction of the sentence from the manuscript is recorded at Addendum 561 of the development's verification ledger; its removal from the module docstring, which the first retraction had missed, at Addendum 563; and its survival in a hand-typed submission form, caught only because the form was read in full before delivery, at Addendum 568. Three removals were required for one sentence, and the third was the one that would have been published.

What is inference, and is labelled as such. We observe that no statement quantified over L for the coupled kernel was attempted in the interval, and we attribute this to the retracted reading continuing to govern target selection. That attribution is ours; it is not established by a diff, and no artefact in the development records a decision not to attempt something. The verifiable half of the observation is the narrower one, and it is enough for this paper: the guards that caught the sentence three times compare identifiers and substrings inside files, so whatever else was carrying the reading was not a file.

3 The chain

Three theorems, each machine-checked, composing into one statement. The first is about a single bond, the second assembles bonds into a matrix, the third is about that matrix and never sees the model.

3.1 The one-bond influence envelope, and that it is least among field-uniform bounds

A two-state site in a real field h has conditional $P(+1) = (1 + \tanh h)/2$, written from the Boltzmann weights and nothing else. Flipping one neighbour across a bond of strength J moves that field to $h - 2J$. Write $\mathrm{tv}(h, h') = |P_h(+1) - P_{h'}(+1)|$.

Theorem 3.1 (machine-checked)**.** *For every $J \geq 0$,*

$$\mathrm{tv}\big(h,\, h - 2J\big) \;\leq\; \tanh J \quad \textit{for all } h, \qquad \textit{and} \qquad \mathrm{tv}\big(J,\, -J\big) \;=\; \tanh J.$$

Hence $\tanh J$ *is the* least *upper bound of* $\{\mathrm{tv}(h, h-2J) : h \in \mathbb{R}\}$.

Proof. $\tanh a - \tanh b = \sinh(a-b)/(\cosh a \cosh b)$, and $\cosh a \cosh b = \big(\cosh(a+b) + \cosh(a-b)\big)/2 \geq \big(1 + \cosh(a-b)\big)/2$ since $\cosh \geq 1$. With $a = h$ and $b = h - 2J$ the difference of the fields is $2J$, so the denominator is at least $(1 + \cosh 2J)/2 = \cosh^2 J$ while the numerator is $\sinh 2J = 2 \sinh J \cosh J$. Dividing by two, $\mathrm{tv} \leq \sinh J \cosh J / \cosh^2 J = \tanh J$. At $h = J$ the two fields are $+J$ and $-J$, their sum is zero, $\cosh 0 = 1$, and every inequality above is an equality. □

What this is, stated carefully, because the obvious reading of it is wrong. The supremum is taken over *all* $h \in \mathbb{R}$. In a concrete finite system with the remaining couplings and any external field fixed, the effective fields actually reachable at a site form a discrete subset of $\mathbb{R}$, and the value $h = J$ at which equality holds need not correspond to any admissible boundary condition of that system. So the true Dobrushin coefficient

$$c_{ij} \;=\; \sup\big\{ \|\mu_i(\cdot \mid \eta) - \mu_i(\cdot \mid \eta')\|_{\mathrm{TV}} \;:\; \eta, \eta' \text{ differ only at } j \big\}$$

of a particular model may satisfy $c_{ij} < \tanh |J_{ij}|$ strictly.

Theorem 3.1 is therefore a **sharp field-uniform one-bond influence envelope**: sharp as a bound over arbitrary real fields, and a valid interdependence majorant, $c_{ij} \le \tanh|J_{ij}|$, for every model with that bond. That is all Dobrushin's condition needs and it invalidates nothing downstream — but it is not the claim that $\tanh|J|$ is the minimal interdependence matrix of any particular system, and this paper does not make that claim. The window below is a sufficient condition, explicit and not optimal.

The attainment still earns its place: without it the envelope would be some number above the influence, and one could not say the envelope is the best field-uniform one. In Lean the three statements are `tvField_le`, `tvField_attained` and `tvField_isLUB`.

3.2 The matrix, and the window as a hypothesis

Given sites ι, bond strengths J and a distance d, put

$$C_{ij} \;=\; \begin{cases} \tanh|J_{ij}| & i \neq j \text{ and } d(i,j) \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 3.2 (machine-checked)**.** *C is entrywise nonnegative, vanishes on the diagonal, and vanishes whenever $d(i,j) > 1$. Consequently* every matrix-side hypothesis *of Theorem 3.3 — (i) and (ii) — holds by construction. Furthermore, if at every site the influence support is contained in a finite set whose bond coefficients sum to at most α, then the row-sum hypothesis holds at every site, with neither the statement nor α mentioning the cardinality of the index type. Finally, at a site carrying two bonds of strength β and two of strength γ,*

$$\sum_j C_{ij} \;=\; 2\tanh|\beta| + 2\tanh|\gamma|.$$

Three remarks, in decreasing order of how easily the statement is over-read.

What is not discharged. The construction settles the matrix-side hypotheses. It does *not* discharge the two conditions on the ambient distance — $d(i,i) = 0$ and the triangle inequality — which Corollary 3.7 still takes as hypotheses, as does the Lean endpoint. For a graph distance they are immediate; for an arbitrary d they are assumptions, and the paper does not pretend otherwise.

One site versus a family. The last display is arithmetic at one site of one five-point graph, and by itself it would say nothing about a family of volumes. The middle clause is what makes it say something: it converts a *local* geometric fact — each site's influence support is covered by a set whose coefficients sum below one — into the row-sum hypothesis holding simultaneously at every site of every volume with that local geometry. In Lean these are `rowSum_bound_of_local_cover` and `dobrushin_bound_of_local_cover`.

The window. It has been carried in this lane's prose as $2\tanh\beta + 2\tanh\gamma < 1$; it is now computed rather than asserted and is consumed by Corollary 3.7. It remains a sufficient condition and, by Section 6, a conservative one. The distance used in the computation is the graph distance on a star, proved reflexive at zero and triangular by exhaustive checking over the 125 triples.

3.3 The resolvent bound

Theorem 3.3 (machine-checked)**.** *Let ι be a finite type, $d : \iota \times \iota \to \mathbb{N}$ satisfy $d(i,i) = 0$ and the triangle inequality, and let $C : \iota \times \iota \to \mathbb{R}$ satisfy* *(i)* $C_{ij} \geq 0$ *for all* i, j*;*

(ii) $C_{ij} = 0$ *whenever* $d(i,j) > 1$*;*

(iii) $\sum_j C_{ij} \leq \alpha$ *for all* i*, with* $0 \leq \alpha < 1$*.*

Then for all i, j *and every* $N \in \mathbb{N}$*,*

$$\sum_{n \in [0,N)} (C^n)_{ij} \;\leq\; \frac{\alpha^{d(i,j)}}{1-\alpha},$$

and consequently $\sum_n (C^n)_{ij} \leq \alpha^{d(i,j)}/(1-\alpha)$*.*

Proof. Three steps, each an induction on n. Nonnegativity is preserved by powers. Row sums multiply: expanding (C^{n+1}) and exchanging the two finite sums, $\sum_j (C^{n+1})_{ij} = \sum_k (C^n)_{ik} \sum_j C_{kj} \leq \alpha \sum_k (C^n)_{ik} \leq \alpha^{n+1}$, so each entry, being at most its row sum, is at most α^n. Support: if $n < d(i,j)$ then $(C^n)_{ij} = 0$, since in $(C^{n+1})_{ij} = \sum_k (C^n)_{ik} C_{kj}$ each k has either $d(i,k) > n$, killing the first factor by induction, or $d(i,k) \leq n$, whence $d(i,j) \leq d(i,k) + d(k,j) \leq n + d(k,j)$ forces $d(k,j) > 1$ and kills the second. Combining, $(C^n)_{ij} \leq \alpha^n$ when $n \geq d(i,j)$ and vanishes otherwise; splitting $[0,N)$ at $d(i,j)$ and reindexing the surviving block by $n = d(i,j) + m$ gives $\alpha^{d(i,j)} \sum_{m < N - d(i,j)} \alpha^m$, and the geometric sum is at most $(1-\alpha)^{-1}$. □

Remark 3.4 (where the volume-freeness is)**.** *The cardinality of* ι *appears nowhere in the hypotheses or the conclusion. That is the entire content of the theorem: the index type may be as large as one likes and neither* α *nor the exponent notices. Every constant is read off single-site data.*

Remark 3.5 (the hypothesis that matters here)**.** *Hypothesis (iii) asks for row sums* bounded *by* α*. It never asks for them to be constant. Proposition 2.2 refutes constancy and leaves (iii) untouched: it is precisely the crack through which the coupled kernel passes after Schur's test has died.*

Proposition 3.6 (non-vacuity, machine-checked)**.** *On the three-point chain with* d *the graph distance and* $C_{ij} = 1/4$ *for* $d(i,j) \leq 1$ *and* 0 *otherwise, hypotheses (i)–(iii) hold with* $\alpha = 3/4 < 1$*, the matrix is not the zero matrix, the two endpoints are at distance exactly two, and the conclusion reads: the whole resolvent series between the endpoints is at most* $9/4$*.*

The witness is not decoration. Hypotheses (ii) and (iii) are jointly satisfiable by the zero matrix, for which the conclusion is trivial, and by data with $d \equiv 0$, for which the exponent is trivial. Proposition 3.6 excludes both.

3.4 The composition

Corollary 3.7 (machine-checked)**.** *Let* J *be bond strengths on a finite set of sites with a distance* d *satisfying* $d(i,i) = 0$ *and the triangle inequality, and let* C *be the matrix of Theorem 3.2. If every row sum of* C *is at most* $\alpha < 1$*, then for all* i, j *and every* N*,*

$$\sum_{n<N} (C^n)_{ij} \;\leq\; \frac{\alpha^{d(i,j)}}{1-\alpha},$$

and the same holds for the full series.

This is the whole chain in one line: the envelope of Theorem 3.1 enters the matrix of Theorem 3.2, which satisfies the hypotheses of Theorem 3.3, whose conclusion mentions no cardinality. In Lean it is `dobrushin_resolvent_bound`, and its proof is the application of one theorem to three others.

4 The measurement, and where the wall is

This section is numerical. Nothing in it is proved, and nothing in it is used by Section 3.

Let $T_L = W_\gamma^{1/2} K_\beta W_\gamma^{1/2}$ be the symmetrised coupled kernel on $\{\pm 1\}^L$ with free spatial boundary, and $r(L) = |\lambda_1|/|\lambda_0|$ its second-eigenvalue ratio, which is specRatio of the development. We compute $r(L)$ for $L = 2, \ldots, 12$ by dense symmetric eigendecomposition and extrapolate by Aitken.

β	γ	$\sinh 2\beta \sinh 2\gamma$	$r(2)$	$r(12)$	$r(\infty)$
controls at zero spatial coupling					
0.20	0.00	0.0000	0.197375	0.197375	$\tanh\beta = 0.197375$
0.40	0.00	0.0000	0.379949	0.379949	$\tanh\beta = 0.379949$
disordered					
0.10	0.10	0.0405	0.1096	0.1209	0.1214
0.20	0.20	0.1687	0.2360	0.2895	0.2922
0.30	0.30	0.4053	0.3733	0.5127	0.5230
0.40	0.40	0.7887	0.5122	0.7813	0.8162
0.10	0.80	0.4783	0.1651	0.4013	0.4570
0.80	0.10	0.4783	0.7246	0.8031	0.8075
ordered					
0.60	0.60	2.2785	0.7489	0.9992	1.000003
0.80	0.80	5.6433	0.8893	0.999999	1.000000

$r(\infty)$ is an Aitken extrapolation of the last three extents. Values above one in the ordered rows are extrapolation overshoot at the sixth decimal and are reported as measured.

Each of the six cells tested inside Onsager's disordered region, $\sinh 2\beta \sinh 2\gamma < 1$, saturates strictly below one, with increment ratios near 0.6; both ordered cells tested reach one to six decimal places. The control at $\gamma = 0$ reproduces $\tanh\beta$ to six decimals, which is Proposition 2.1 — the harness is calibrated against a theorem of the development rather than against itself.

What these data support, stated at their strength and no further. They are consistent with the degeneracy being a property of the *ordered* phase, and they are *not* consistent with the degeneracy being caused by switching on the spatial weight: six disordered cells with the weight switched on fail to degenerate. That second, negative reading is the one this section is for, and it is the one that reopened the question.

What they do not support. They do not locate the transition. Eight cells at $L \leq 12$, with a three-term Aitken extrapolation whose overshoot is visible in the ordered rows above, cannot place a phase boundary; doing so would require a dense sweep near $\sinh 2\beta \sinh 2\gamma = 1$, finite-size scaling, and control of the extrapolator, none of which is attempted here. A reader should take from this section that the wall is *not* where the development had been treating it as being, and nothing about where it is.

Gates. Three gates were registered in `scripts/judge_dobrushin.py` together with the lane charter, before a line of the Lean of Section 3 was written. Each exits non-zero on failure; none bundles a theorem with its witness; the bound gate predicts a number rather than sampling. One of them was later found not to exercise the hypothesis of Remark 3.5 — it normalised every row to *exactly* α — and was redesigned to carry unequal row sums, strictly harder, declared in its own docstring and committed before being run. The gates emit a verdict only after an explicit counter confirms that the expected number of checks actually executed, because an empty failure list is also what zero checks produce. No acceptance decision anywhere in this lane depends on a Python `assert`.

5 What remains, named exactly

Theorems 3.1–3.3 are an engine, not a conclusion. Turning them into a volume-uniform statement about specRatio requires, in order:

1. a valid single-site interdependence majorant, and the sufficient window it produces. **Done**: Theorems 3.1 and 3.2, machine-checked, with the window computed rather than asserted, and holding at every site of every volume with the given local geometry. Not done, and not needed: the *minimal* interdependence matrix of any particular system;
2. Dobrushin's comparison estimate [2, 3] — that the connected correlations of the Gibbs measure are controlled by the resolvent of C, hence decay exponentially at a rate free of the extent. **Open, and this is the bottleneck**;
3. transport of that decay into the operator formulation.

Step 2 is the only *new analytic estimate* currently identified, and nothing in this paper reduces its difficulty. It is not the only remaining work: the downstream consumer exists but its instantiation still needs the finite-time interface described below, so "one obligation left" would be too tidy a sentence for the actual state. One structural remark, offered as orientation and not as progress: in this model the configuration space of a finite system is finite, so the conditionals are ratios of finite sums and the comparison estimate does not require the measure-theoretic apparatus in which it is usually stated. Whether that makes it tractable is not established here.

The consumer, named. It is `volumeUniform_gap` in `YangMills/OS/TransferGap.lean`, and its signature, read off the elaborated declaration rather than from memory, is: for a family of real inner-product spaces H_i indexed by ι, transfer operators T_i and vectors Ω_i with `VacuumTransfer` $T_i\ \Omega_i$ for every i, and a rate $0 < r < 1$, the hypothesis

$$\forall i,\ \forall v \in H_i,\ \exists C,\ \forall n, \quad \left|\texttt{connCorr}\, T_i\, \Omega_i\, v\, n\right| \ \le\ C\, r^{\,n}$$

yields $\exists m > 0$ with $\|\texttt{projectedTransfer}\, T_i\, \Omega_i\| \le e^{-m}$ for every i. It takes, over a family of finite volumes, decay at a *common* rate and returns a *common* mass. Its quantifier order is worth stating because it is load-bearing: the prefactor C may depend on the volume and on the observable, and only r is shared. The development's own standing objection — that a bound $C(L)\rho^N$ with $C(L) \to \infty$ says nothing — therefore does not apply to it, the conclusion being an operator-norm bound.

A design decision is recorded here rather than discovered later: the vacuum pairing is an infinite-time object, so step 3 must proceed through finite time extent with explicit boundary control, taking the limit only at the level of the bound. No infinite-volume state is constructed anywhere in this paper, and the word *clustering* is accordingly not used for one.

6 Prior art, and the exact delta

Dobrushin's uniqueness theorem is classical [1, 2, 3] and nothing in this paper is new mathematics. We locate the contribution precisely.

Not explicit constants. The cell here is the anisotropic two-dimensional Ising model, whose transfer spectrum has been available in closed form since Onsager [4] and Kaufman [5]. A constant obtained from Dobrushin's condition is *worse* than the exact answer — on the cells we measured, by a factor between roughly two and thirty-six. Any claim to have improved a constant for this model would be false, and we make none.

Generality, instead. Dobrushin's coefficient is read off the single-site conditionals of an *arbitrary* positive finite-range slice weight, which is what the development's Gibbs weight quantifies over, and where the exact solution is silent.

Mechanisation. A search conducted on 1 August 2026 over the public proof-assistant repositories and the arXiv listings for Lean, Coq and Isabelle returned no mechanisation of the finite-lattice Dobrushin interdependence matrix or of the comparison estimate. We state it that narrowly on purpose: the name *Dobrushin* attaches to several distinct results, and mechanised work exists that uses others of them, so a blanket claim of first formalisation would be both unsupported by the search we ran and ambiguous about which theorem it referred to. The same search does return a Lean 4 formalisation of the transverse-field Ising chain [9], with momentum-mode decomposition and a Jordan–Wigner transformation; it concerns the quantum chain rather than the classical two-dimensional transfer matrix, and displaces nothing here, but it means a blanket claim that the Ising model has not been formalised would not survive review, and we make none. A formalisation of the Perron–Frobenius theorem for irreducible nonnegative matrices also exists [8], as does a mechanisation of the Osterwalder–Schrader axioms for the Gaussian free field [10]. Reflection positivity and finite-lattice spectra for related models have been treated without mechanisation [11].

7 Limitations

All three theorems are elementary. Theorem 3.3 is three inductions; Theorem 3.1 is a hyperbolic identity and a bound on a denominator; Theorem 3.2 is a construction and a finite computation. Their value is that they are machine-checked, non-vacuous, composed, and carry the hypothesis the dead method violated — not that any of them is deep.

Section 4 is numerics at $L \leq 12$ with an Aitken extrapolation. It refutes an attribution; it does not prove where the wall is, and an extrapolation can mislead near a transition. The window is not optimal and is never called sharp.

Step 2 of Section 5 is open, and until it closes the volume-uniform statement about specRatio for the coupled kernel remains unproved. This is the substantive limitation and the chain above does not shorten it: what the chain does is remove every *other* obligation from between the envelope and the resolvent bound, so that a reader can see that one thing is missing and what it is.

8 Reproducibility

Anchor. All sources cited above are read at commit `c3d8e32d2eecf85cf2dba26ce9b0a8bb42005c9c` of `https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME`. Toolchain `leanprover/`

`lean4:v4.29.0-rc6`; Mathlib pinned at `07642720480157414db592fa85b626dafb71355b`. A runner verifies all three of these before compiling anything and aborts on any mismatch, so that a stale cache cannot masquerade as a reproduction.

Job counts, and how each delta was obtained. The development machine measured the core at 8465 jobs with Theorem 3.3's module removed from the root and 8466 with it restored — that delta by rebuilding the same tree twice, not by subtracting against a baseline someone else measured. A fresh Linux clone reproduced 8466 at that commit. With the modules of Theorems 3.1 and 3.2 added, the Linux plane measures 8468 with zero errors: the +2 is the difference between two measurements on the same plane, one job per new module, and is not an extrapolation. At the anchor the count is still 8468, because the last four theorems were added to an *existing* module and a declaration does not create a build job — only a module does.

Oracle. At the anchor the repository-wide oracle ran to completion with exit code zero over 2817 axiom reports and **zero** occurrences of `sorryAx`. The accounting, spelled out because a count of *newly added* declarations is not a count of the paper's modules: twenty-three new oracle reports come from the two modules added over the last two anchors — eleven for Theorem 3.1, twelve for Theorem 3.2, of which four were added after an external reading identified the gap between a single site and a family of volumes — while the fourteen declarations of Theorem 3.3's module, present since the first anchor, are also in the oracle and were separately checked one by one on the development machine. Every one of the thirty-seven depends on exactly `[propext, Classical.choice, Quot.sound]`: Lean's three standard axioms, no project axiom, and no `sorry` anywhere.

One gap in the runner's own coverage, stated because a reader counting modules would otherwise not find it. The runner builds two lane modules by name; `DobrushinRowSum.lean` is not one of them. It is nevertheless imported by the core, so the zero exit of the core build covers it, and its declarations appear in the oracle.

Independent hash agreement. A second desk, working from its own check-out on a different operating system and without coordination on the value, reported `12c1fcdb12e39e46381fa372a102503f5d63522ecbae27c0b3911671a7c04eb7` at 11467 bytes for Theorem 3.3's module. The Linux run computed the same digest and the same size. The oleans, the axiom reports and the elaboration therefore refer to the same bytes, which is the only reason any of the numbers above can be compared at all. At the anchor the two further modules hash to `fe16a382e297c6f1e668677d54117eb7873e03ab981818cbea123bfa50af47c9` and `68bb7408ebe7af782804238a4780fcebee499bdb512b3435703ea8f7c1a69cf2`.

Certifiers. Both gate scripts were run in `normal` and in `optimized` (`python -O`) mode on the Linux plane. All four runs exited zero: `judge_dobrushin.py` reported 29 of 29 checks in each mode, and `judge_dobrushin_d2.py` reported 18 of 18. No acceptance decision in either script depends on a Python `assert`; each counts the checks it performed and refuses to print a verdict when fewer ran than expected, because an empty failure list is also what zero checks produce.

Child processes. Every child of the runner writes a sentinel holding exactly one line, its real decimal exit code, captured after termination and written atomically with validation. Four states are distinguished on read: absent, malformed, non-zero, zero. A run in which every sentinel reads zero is reported as a *pass candidate*, never as a pass, because the logs still have to be read.

Every module in the anchor compiles. An earlier version of this paper was anchored to a tree in which `DobrushinCoefficient.lean` did not, and said so in this section rather than leaving it to be discovered. It took five passes to close, and the record of those passes is worth one sentence because it is the honest description of what mechanising a classical argument costs: none of the fifteen errors across them was mathematical. The mechanism — the identity of Theorem 3.1 and the denominator bound under it — is the one written in the first draft and was never altered. Every failure was a library lemma whose name or form had been recalled wrongly, or a tactic emitting a different number of goals than the script assumed.

References

[1] R. L. Dobrushin, *The description of a random field by means of conditional probabilities and conditions of its regularity*, Theory of Probability and its Applications **13** (1968), 197–224. The uniqueness condition and the comparison estimate whose mechanised form is step 2 of Section 5. The theorem of this paper is classical and due here.

[2] R. L. Dobrushin, *Prescribing a system of random variables by conditional distributions*, Theory of Probability and its Applications **15** (1970), 458–486.

[3] B. Simon, *The Statistical Mechanics of Lattice Gases, Volume I*, Princeton University Press, 1993. Chapter V gives the Dobrushin uniqueness condition, the interdependence matrix, and the exponential decay of correlations it yields; the one-bond envelope $\tanh |J|$ for the Ising cell and the resulting window $2 \tanh \beta + 2 \tanh \gamma < 1$ are the standard specialisation.

[4] L. Onsager, *Crystal statistics I: a two-dimensional model with an order-disorder transition*, Physical Review **65** (1944), 117–149. The exact solution, and the source of the line $\sinh 2\beta \sinh 2\gamma = 1$ used as the reference boundary in Section 4.

[5] B. Kaufman, *Crystal statistics II: partition function evaluated by spinor analysis*, Physical Review **76** (1949), 1232–1243. The transfer-matrix spectrum in closed form. This is the reason Section 6 claims generality rather than constants: for this cell the exact second eigenvalue is available and the Dobrushin constant is worse than it.

[6] I. Schur, *Bemerkungen zur Theorie der beschränkten Bilinearformen mit unendlich vielen Veränderlichen*, Journal für die reine und angewandte Mathematik **140** (1911), 1–28. The test whose constant-row-sum form is the method discussed in Section 2.

[7] E. Seneta, *Non-negative Matrices and Markov Chains*, revised printing, Springer, 2006. Standard reference for nonnegative matrices, ergodicity coefficients and the resolvent series $\sum_n C^n = (I - C)^{-1}$ of which Theorem 3.3 is a locally-supported, entrywise refinement.

[8] The mathlib Community, *The Lean mathematical library*, Proceedings of CPP 2020, 367–381. The library against which the development is pinned; it carries a formalisation of the Perron–Frobenius theorem for irreducible nonnegative matrices.

[9] *A Machine-Verified Proof of a Quantum-Optimization Conjecture*, arXiv:2606.29687 (2026). A Lean 4 formalisation of the Ising chain with momentum-mode decomposition and the Jordan–Wigner transformation. It concerns the transverse-field quantum chain rather than the classical two-dimensional transfer matrix, and displaces nothing here; it is cited because a blanket claim that the Ising model has not been formalised would be false.

[10] *Formalization of Quantum Field Theory*, arXiv:2603.15770 (2026). Osterwalder–Schrader axioms for the Gaussian free field in Lean 4; the nearest mechanised neighbour on the constructive side, and free rather than interacting.

[11] *Spectrum and reflection positivity on a finite lattice*, arXiv:2606.07922 (2026). Finite-lattice spectral and reflection-positivity results for a related model, not mechanised.